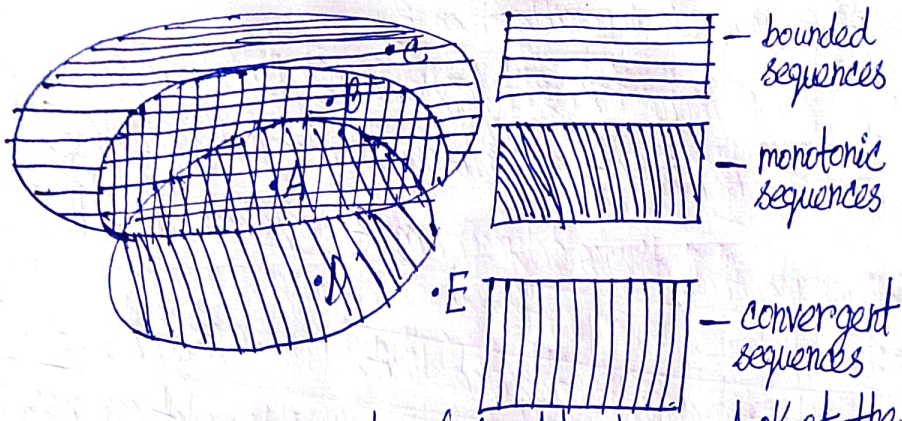

Author: In order to avoid confusion, let us have a look at the above illustration. Let us assume that all bounded sequences are "collected" (as if we were picking marbles scattered on the floor) in an area shaded by horizontal lines, all monotonic sequences are collected in an area shaded by titted lines, and, finally, all convergent sequences are collected in an area shaded by vertical lines.

- As follows from the figure, the area shaded vertically is completely included into the area shaded horizontally. It means that any convergent sequence must be also bounded.
- The overlapping of the areas shaded horizontally and by titted lines occurs inside the area shaded vertically. It means that any sequence that is both bounded and monotonic must be convergent as well.
- It is easy to deduce that only five types of sequences are possible. In the figure the points designated by A, B, C, D and E identify five sequences of different types. Try to name these sequences and find the corresponding examples among the sequences discussed in Dialogue 1.

Sam: A → Bounded and monotonic. ∴ Convergent.
Seq 5, Seq 7, Seq 13 are A.

B → Bounded and Convergent. But not monotonic
Seq 9 is B

C → Bounded. Not Convergent, not Monotonic
Seq 10 is C

D → Monotonic. Not Convergent, not Bounded
Seq 1, 2, 3, 4, 6, 12, 11 are D

E → Not Bounded, Not Monotonic, Non Convergent
Seq 8 is E